Visualizing genus zero dessins

Mike Musty ICERM 2025 August 12, 2025

Intro

Background

- Number theory, computation

Past

- Postdoc (ICERM)
- Mathematician (CRREL)
  - Computer vision, mixed models

Current

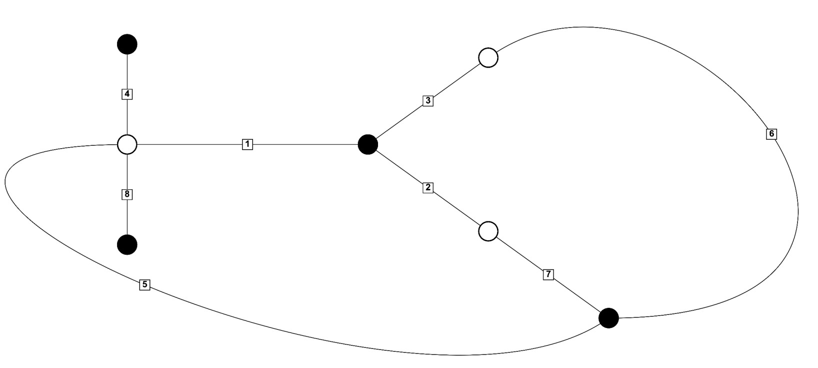
- Data wrangler (Faraday)
  - Product implementation
  - Modeling / API infrastructure
  - Client solutions
- Toddler wrangler

Outline

1. Drawings
2. Background
3. Visualizations old and new
4. Examples and motivating questions

Drawing example

σ_0 = (1,4,5,8)(2,7)(3,6), σ_1 = (1,2,3)(5,6,7), σ_∞ = (8,7,1)(6,2)(5,4,3)
Base field: 1 - 3T^2
Embedding: -1.732051



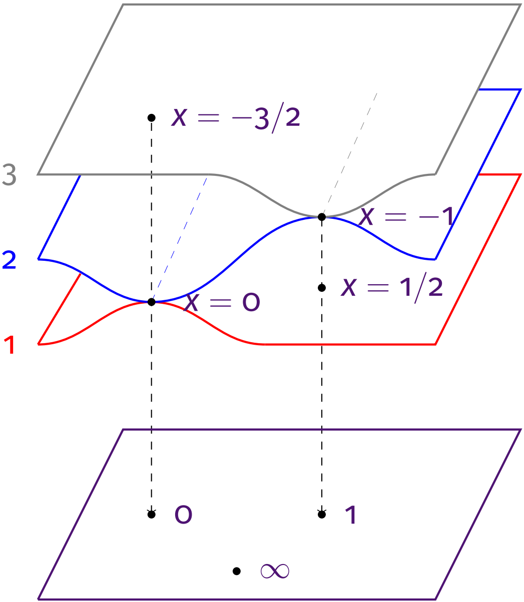
8T38-4.2.2_3.3.1.1_3.3.2

Degree 3 example

$$\big((1\,2)(3),\,(1)(2\,3),\,(1\,3\,2)\big)$$

$$\Big\updownarrow$$

Degree 3 example



X

$\downarrow$

$\mathbb{P}^1$

$$\varphi(x) = \underbrace{2x^3 + 3x^2}_{z} = x^2(2x+3) = (2x-1)(x+1)^2 + 1$$

Belyi maps, dessins, etc.

- This branched covering is an example of a **Belyi map**, a map of Riemann surfaces unbranched outside of $0, 1, \infty$.
- A Belyi map determines a **permutation triple** via monodromy.
- A permutation triple determines a **triangle subgroup** Γ which acts on a spherical, Euclidean, or hyperbolic space $\mathcal{H}$ such that $\Gamma\backslash\mathcal{H} \to \Delta\backslash\mathcal{H}$ is the corresponding Belyi map.
- The preimage of $[0, 1]$ is a (connected) bipartite graph embedded on a surface called a **dessin d'enfant**.

Equivalent categories

$$
\begin{tikzcd}[column sep=small]
\left\{\begin{array}{l}\text{Transitive permutation triples}\\ (\sigma_0,\sigma_1,\sigma_\infty)\in S_d^3 \text{ with } \sigma_\infty\sigma_1\sigma_0 = \\ 1 \text{ up to simultaneous conjuga-}\\ \text{tion}\end{array}\right\} \arrow[dr, leftrightarrow, "\sim"] & & \left\{\begin{array}{l}\text{Conjugacy classes of in-}\\ \text{dex } d \text{ triangle subgroups}\\ \Gamma \leq \Delta(a,b,c)\end{array}\right\} \\
 & \left\{\begin{array}{l}\text{Isomorphism classes of}\\ \text{dessins with } d \text{ edges}\end{array}\right\} \arrow[ur, leftrightarrow, "\sim"] \arrow[dl, leftrightarrow, "\sim"] \arrow[dr, leftrightarrow, "\sim"] & \\
\left\{\begin{array}{l}\text{Isomorphism classes of degree } d \text{ function}\\ \text{field extensions } K \supseteq \mathbb{C}(z) \text{ with discrimi-}\\ \text{nant supported above } 0,1,\infty\end{array}\right\} & & \left\{\begin{array}{l}\text{Isomorphism classes of}\\ \text{Belyi maps of degree } d\end{array}\right\}
\end{tikzcd}
$$

Degree 3 example

$$\Big((1\,2)(3), (1)(2\,3), (1\,3\,2)\Big)$$

$$[\Delta(2,2,3) : \Gamma] = 3$$

$$\mathbb{C}\Bigg(\underbrace{2x^3 + 3x^2}_{z}\Bigg) \subseteq \mathbb{C}(x)$$

$$\varphi(x) = \underbrace{2x^3 + 3x^2}_{z}$$

Galois action

Theorem (Belyi, 1979)

A smooth projective curve X over $\mathbb{C}$ can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a nonconstant morphism of algebraic curves $\varphi : X \to \mathbb{P}^1$ unramified outside $\{0, 1, \infty\}$.

In the 1980s Grothendieck described an action of the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on the set of (isomorphism classes of) dessins.

Passports

A **passport** consists of the data (G, λ) where

- $G \leq S_d$ is a transitive subgroup
- $\lambda = (\lambda_0, \lambda_1, \lambda_\infty)$ is a triple of partitions of d

A permutation triple $\sigma \in S_d^3$ **belongs** to (G, λ) if

- $\langle \sigma_0, \sigma_1, \sigma_\infty \rangle = G$
- $\sigma_0, \sigma_1, \sigma_\infty$ have the cycle types specified by $\lambda_0, \lambda_1, \lambda_\infty$

Note

- The **size** of a passport is the number of permutation triples belonging to it (up to simultaneous conjugacy).
- A **refined passport** replaces λ with $C = (C_0, C_1, C_\infty)$ a triple of conjugacy classes of G.
- Passports and refined passports are stable under the Galois action and therefore are unions of Galois orbits.

Objective of this work

Draw topologically correct dessins to help visualize the Galois orbits of all (genus zero) passports in the LMFDB.

- `https://www.lmfdb.org/Belyi/`
- https://michaelmusty.github.io/dessins/

7T6-5.1.1_4.2.1_3.2.2

- https://www.lmfdb.org/Belyi/7T6/5.1.1/4.2.1/3.2.2/

- https://michaelmusty.github.io/dessins/passports/7T6-5.1.1_4.2.1_3.2.2/index.html

- All drawings together

5T4-5_5_3.1.1

- https://www.lmfdb.org/Belyi/5T4/5/5/3.1.1/
- https://michaelmusty.github.io/dessins/passports/5T4-5_5_3.1.1/index.html
- All drawings together

9T34-6.1.1.1_5.2.2_6.1.1.1

- `https://www.lmfdb.org/Belyi/9T34/6.1.1.1/5.2.2/6.1.1.1/`
- `https://michaelmusty.github.io/dessins/passports/9T34-6.1.1.1_5.2.2_6.1.1.1/index.html`
- All drawings together

Motivating future questions

The ultimate goal is to learn something about the absolute Galois group through its action on dessins. To that end:

- Can we tell when 2 dessins correspond to Galois conjugate Belyi maps?
- Is there topological or combinatorial data that can tell us when a passport splits into multiple Galois orbits?
- Can we determine the Galois orbits given a set of representatives for a passport?

Completeness of the data

d \ g	0	1	2	3	≥ 4	total
1	1/1	0	0	0	0	1/1
2	1/1	0	0	0	0	1/1
3	2/2	1/1	0	0	0	3/3
4	6/6	2/2	0	0	0	8/8
5	12/12	6/6	2/2	0	0	20/20
6	38/38	29/29	7/7	0	0	74/74
7	89/89	50/50	7/13	2/3	0	148/155
8	243/261	83/217	0/84	0/11	0	326/573
9	410/583	33/427	0/163	0/28	0/6	443/1207
total	802/993	204/732	16/269	2/42	0/6	1024/2042

Thanks!

To:

- Sam Schiavone
  `https://math.mit.edu/~sschiavo/`
- Aaron Fenyes `https://ooo.fareycircles.ooo/`
- And to you for listening!